

Strings

String is a collection of characters. C++ provides two types of String representations:

- The C-style String (Character Array)
- Strings that are objects of string class

The C-Style String (Character Array)

The string can be defined as the one - dimensional array of characters terminated by a null ('\0').

Declaration of string:

Syntax: `char stringname[size];`

Ex. `char s[20];`

String Initialization:

`Char s[6]={ 'h', 'e', 'l', 'l', 'o', '\0' };`

`Char s[]="hello";`

Note: computer automatically appends a null character('\0') at the end of the string.

The size of string should be no of characters plus 1.

`char name[6]={ 'r', 'a', 'h', 'u', 'l', '\0' };`

`char name[]="rahul";`

Here the string "rahul" contains 5 characters so compiler automatically allocates 6 memory locations (1 extra for storing NULL character).

Print String

```
for(i=0; name[i] != '\0'; i++)
{
    cout << name[i];
}
```

Reading & Writing of String using Character Array

`char name[12];`

`cin >> name;`

```

cout << name;
#include <iostream>
using namespace std;
int main ()
{
    char greeting[6] = {'H', 'e', 'l', 'l', 'o', '\0'};
    cout << "Welcome message: ";
    cout << Hello << endl;
    return 0;
}

```

OUTPUT

Welcome message: Hello

```

#include <iostream>
using namespace std;
int main()
{
    char str[100];
    cout << "Enter a string: ";
    cin >> str;
    cout << "You entered: " << str << endl;
    cout << "\nEnter another string: ";
    cin >> str;
    cout << "You entered: " << str << endl;
    return 0;
}

```

OUTPUT

Enter a string: C++

You entered: C++

Enter another string: Programming is fun

You entered: Programming

Note: In the second example, only “Programming” is displayed instead of “Programming is fun”. This is because the extraction operator >> considers a space “ ” as a terminating character.

C++ program to read and display an entire line entered by the user.

```
#include <iostream>
using namespace std;
int main()
{
    char str[100];
    cout << "Enter a string: ";
    cin.getline(str, 100);
    cout << "You entered: " << str << endl;
    return 0;
}
```

OUTPUT

Enter another string: Programming is fun

You entered: Programming is fun

- To read the text containing the blank space, cin.getline function can be used. This function takes two arguments.
- The first argument is the name of the string, and the second argument is the size of the array.
- getline() will take input till it gets a new line(\n) character.

Library Function in <cstring>	Use	Description

strlen()	strlen("hello"); or strlen(s);	computes string's length
strcpy()	strcpy(s1,"hello"); or strcpy(s1,s2);	copies a string to another
strcat()	strcat(s1, s2); or strcat(s1, "hello");	Concatenates (joins) second argument to first argument
strcmp()	strcmp(s1, s2); or strcmp(s1, "hello"); or strcmp("hello", s2); or strcmp(s1, "hello");	compares two strings s1, s2 and if they matched returns 0 otherwise returns non zero value.

String Handling Functions

```
#include<iostream>
#include<cstring>
using namespace std;
int main()
{
    char f_name[20];
```

```

    char l_name[20];
    char full_name[50];
    cout<<"Enter your first name: ";
    cin>>f_name;
    cout<<"Enter your last name: ";
    cin>>l_name;
    cout<<"Hi " <<f_name<<"! Your first name has "<<strlen(f_name)<<"
characters."<<endl<<"And your last name has "<<strlen(l_name)<<"
characters."<<endl;
    strcpy(full_name,f_name);
    strcat(full_name, " ");
    strcat(full_name,l_name);
    cout<<"Your Full name is: "<<full_name<<endl;
    if((strcmp(f_name,l_name))==0)
        cout<<"Both strings are equal."<<endl;
    else
        cout<<"Strings are not equal."<<endl;
    return 0;
}

```

OUTPUT

Enter your first name: John

Enter your last name: Winson

Hi John! Your first name has 4 characters.

And your last name has 6 characters.

Your Full name is: John Winson

Strings are not equal.

Note: For using C-style character strings, <cstring> header must be used.

std::String class

The std::string class provides a convenient way to handle strings in C++. It is part of <string> header.

Declaration and Initialization of string:

Syntax:

```
#include <string>

std::string str = "Welcome";
```

OR

```
std::string str = {"Welcome"};
```

OR

```
std::string str {"Welcome"};
```

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    //Declaring a string object
    string str;
    cout << "Enter a string: ";
    getline(cin,str);
    cout << "You entered: " << str << endl;
    return 0;
}
```

OUTPUT

Enter a string: Programming is fun

You entered: Programming is fun

- getline() will take input till it gets a new line(\n) character.
- getline() function takes the input stream as the first parameter, which is cin and str as the location of the line to be stored.

Library Function in <string>	Description
find()	Find a substring in the string.
rfind()	the last occurrence of a substring in the string.
append()	Append to the string.
insert ()	Append to the string.
erase ()	Erase characters from the string.
replace ()	Replace portions of the string.
compare ()	Compare two strings.
Library Function in <string>	Description
substr()	Extract a substring from a string.
length() or size()	Returns the length of the string
at() or using array index	To access individual characters using array indexing

C_str()	To obtain a C-style string from a std::string.
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```
#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = {"hello"};
    string str2 {"Joy"};
    string str3 {str2};
    string str4 (5,'a'); //Displaying 'a' 5 times
    cout<<str1<<endl<<str2<<endl<<str3<<endl<<str4;    return 0;
}
```

OUTPUT

Hello

Joy

Joy

aaaaa

```
#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = {"hello"};
    string str2 {str1,3}; //Displaying characters    from
index 3 (starting from 0)
    string str3 {"Dipika",3}; //Displaying first 3
    characters
```

```

        cout<<str1<<endl<<str2<<endl<<str3;
        return 0;
}

```

OUTPUT

```

Hello
lo
Dip

```

Example : C++ program to accessing a character from a string

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str = {"Welcome Home"};
    cout << str[6]<<endl; //Displays character at index 6. No Bound
Checking
    cout << str.at(6); //Displays character at 6th index
    return 0;
}

```

OUTPUT

```

e
e

```

Example : C++ program to print the string using range based for loop

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str = {"Welcome Home"};

```

```

        for(char c:str)
        {
            cout<<c;
        }

        return 0;
    }

```

OUTPUT

Welcome Home

Example: C++ program to concatenate two strings and display the length of a string

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = "Welcome";
    string str2 = "Home";
    cout<<str1+str2<<endl; //Concatenates two strings
    cout<<str1.append(str2); //Append str1 with str2
    cout<<str1+" "+str2+ " "+"Joy"<<endl;
    cout<< "Length of str1: "<<str1.length()
    return 0;
}

```

OUTPUT

WelcomeHome

WelcomeHome

Welcome Home Joy

Length of str1: 7

Note: For length of the string, size() function can also be used.

Example : C++ program to compare strings

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = "Welcome";
        string str2 = "Home";          cout<<(str1==str2)<<endl;
//Compares whether the Strings are same or not
        cout<<(str1<str2)<<endl; // Checks the ASCII Value of the
First Character
        return 0;
}

```

OUTPUT

0

0

Example: C++ program to compare strings using compare() function

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = "Welcome";
        string str2 = "Home";
        if ((str1.compare(str2)) == 0)
            cout << "String Matched" << endl;
        else
            cout << "String Not Matched" << endl;
        return 0;}

```

OUTPUT

String Not Matched

Example: C++ program to display the substring of a string

```
#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = "Welcome Home";
    cout<<str1.substr(3,2)<<endl; //Display the substring from
index 3. Length of substring is 2.
    return 0;
}
```

OUTPUT

co

C++ program to find the substring of a string

```
#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str = "Hi! Welcome Home, Joy. Hi";
    cout<<str.find("Jo")<<endl;      cout<<str.find("Jo",19)<<endl;
    cout<<str.rfind("Hi");
    return 0;
}
```

OUTPUT

18

18446744073709551615

23

Example: C++ program of inserting into a string

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str = "Hi! Welcome Home";
    string str1 = "I am Nik";
    cout<<str.insert(4, "Joy, ")<<endl;
    cout<<str.insert(4,str1)<<endl;          cout<<str.insert(21, ".
What will you
                                     have?",4,5);

return 0;
}

```

OUTPUT

```

Hi! Joy, Welcome Home
Hi! I am NikJoy, Welcome Home
Hi! I am NikJoy, Welcat wiome Home

```

Example: C++ program of Replacing a string using replace() function

```

#include <iostream>
#include<string>
using namespace std;
int main()
{
    string str1 = "Hey World";
    string str2 = "Hello";
    str1.replace(0, 3, str2); // Replaces 3 characters from 0th index
by str2
    cout << str1 << endl;
    return 0;
}

```

```
}
```

OUTPUT

Hello World

Example: C++ program of erasing a character from a string using erase() function

```
#include <iostream>
```

```
#include<string>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    string str = "Hello, World";
```

```
    // Erase a single character starting from index 5
```

```
    str.erase(5, 1);
```

```
    cout << str;
```

```
    return 0;
```

```
}
```

OUTPUT

Hello World